



Time Identification of Non-Commensurable Fractional Order Systems of the Second Kind

This is the Thesis Subtitle if necessary.

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Mechanical Engineering

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

I would like to thank my parents for their friendship, encouragement and caring over all these years, for always being there for me through thick and thin and without whom this project would not be possible. I would also like to thank my grandparents, aunts, uncles and cousins for their understanding and support throughout all these years.

To each and every one of you – Thank you.

Abstract

lorem ipsum

Keywords

Maecenas tempus dictum libero; Donec non tortor in arcu mollis feugiat;Cras rutrum pulvinar tellus.

Resumo

lorem ipsum

Palavras Chave

Colaborativo; Codificação; Conteúdo Multimédia; Comunicação;

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Acronyms

FOTF	Fractional-Order Transfer Function
HD	High Definition
SD	Standard Definition
WLAN	Wireless Local Area Network
WWAN	Wireless Wide Area Network

1

Introduction

2

State of the Art

3

Mathematical Modeling of Fractional Order Systems

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3.1 General Fractional-Order Transfer Functions

Many real-world phenomena, such as anomalous diffusion and viscoelasticity cannot be easily modeled by integer-order equations. Fractional Calculus tackles this problem by expanding the concepts of differentiation and integration to non-integer orders that better represent the systems mentioned before.

This mathematical extension leads to the definition of the Fractional-Order Transfer Function (FOTF), where integer powers of the Laplace variable s are replaced by real-valued exponents.

A general fractional-order system can be represented by a transfer function like the following:

$$G(s) = \frac{b_m s^{\beta_m} + b_{m-1} s^{\beta_{m-1}} + \dots + b_0 s^{\beta_0}}{a_n s^{\alpha_n} + a_{n-1} s^{\alpha_{n-1}} + \dots + a_0 s^{\alpha_0}} \quad (3.1)$$

where α_k, β_k are non-negative real numbers. These exponents provide an additional degree of freedom for modeling complex dynamics.

3.1.1 Commensurate vs. Non-Commensurate Systems

Fractional Order systems are categorized by the relationship between their differentiation orders, *i.e.*, the values of the transfer function's exponents:

1. **Commensurate Order Systems:** A system is said to be commensurate order if all the exponents ν_k are integer multiples of a single base order ν .
2. **Non-Commensurate Order Systems:** A system is non-commensurate when its exponents cannot be expressed as multiples of a common base order.

3.2 The Second Species Model

This research will be focused on a specific class of fractional-order models known as "second species" systems, since they resemble second-order transfer functions. In this case, the differential orders are ν and $\nu + 1$ as shown below:

$$G(s) = \frac{K e^{-\theta s}}{1 + 2\zeta \left(\frac{s}{\omega_n}\right)^\nu + \left(\frac{s}{\omega_n}\right)^{\nu+1}} \quad (3.2)$$

where the parameters are defined as follows:

- K : Gain of the system;
- $e^{-\theta s}$: delay time;

- $\nu \in \mathbb{R}^+$: the fractional order of the system;
- ζ : the pseudo-damping coefficient;
- ω_n : the natural frequency of the system.

3.2.1 Structural Analysis

This model is classified as non-commensurate since the powers of the Laplace variable s (in this case, ν , $\nu + 1$) are not necessarily multiples of a single base order unless ν itself is an integer, which defeats the purpose. This structural choice is important because it ensures the natural frequency ω_n acts as a scaling factor in time, as will be proved in the next sections.

When $\nu = 1$, the transfer function reverts to the classical second-order normalized form:

$$G(s) = \frac{K e^{-\theta s}}{1 + 2\zeta \left(\frac{s}{\omega_n}\right) + \left(\frac{s}{\omega_n}\right)^2} = \frac{K \omega_n^2 e^{-\theta s}}{s^2 + 2\zeta \omega_n s + \omega_n^2} \quad (3.3)$$

This backwards compatibility ensures the identification model will hold its value for traditional integer-order systems while expanding their reach to fractional orders.

3.3 Stability Analysis of the non-commensurate system

The transition from integer-order to fractional-order systems introduces significant complexity in determining the stability criteria. Classical methods like the Routh-Hurwitz criteria become highly complex for cases such as the system studied by this article [1].

3.3.1 Stability Criteria

The work done by *Hmed et al.* [1] treats the plant as a double nested loop and assesses the stability using the Nyquist Criteria. This research's starting point will be fully based on the stability conditions found in that article. In this case, the system is stable if, and only if, the following conditions apply:

- $0 < \nu \leq 1$ and $\zeta \geq 0$;
- $0 < \nu < 1$, $\zeta < 0$ and $\frac{-2^{\nu+1}\zeta}{\cos(\frac{\nu\pi}{2})} (-\zeta \tan(\frac{\nu\pi}{2}))^\nu < 1$;
- $1 < \nu < 2$, $\zeta > 0$ and $(2\zeta\omega_n)^2\omega_u^{2\nu} + \omega_u^{2\nu+2} - \omega_n^{2\nu+2} > 0$ with $\omega_u = 2\zeta\omega_n \tan\left(\frac{(2-\nu)\pi}{2}\right)$,

which result in the Stability Diagram presented in Figure 3.1.

Check
equations
in article

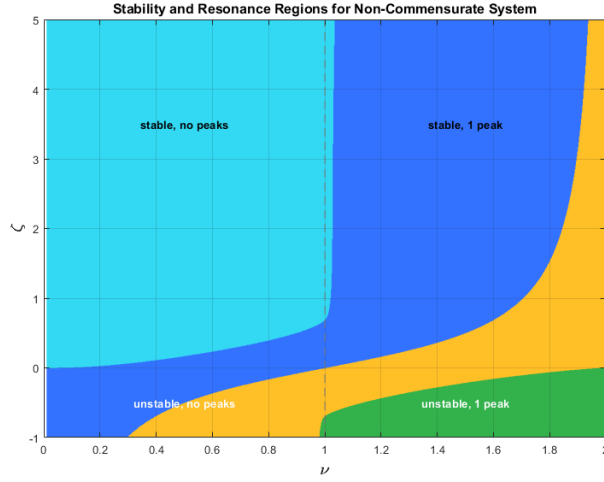


Figure 3.1: Stability Chart of the Fraction Order Transfer Function

3.4 Problem Statement and Goals

Despite the analytical progress made in characterizing the stability and resonance of the fractional order system, as seen in the work of *Hmed et al.* [1] and *Valério et al.* [2,3], there is still a challenge to be tackled: the identification of such systems from experimental or simulated data.

While identification for integer-order systems is a mature field with well-established methods, the nature of fractional systems require a different approach. This dissertation will aim to address this by developing an empirical Identification Model that uses extracted parameters from the unit step time response of said systems.

3.4.1 Formulation of the Problem

Let's consider that a step with amplitude L is applied to the input of the system described by Equation (3.2). From its output, $y(t)$, we know that:

- The system's gain K can be found as $\frac{1}{L} \lim_{t \rightarrow \infty} y(t)$ [3];
- The delay time θ can be found by computing the first time instant that verifies $y(t) \neq 0$. Although some issues may arise when analyzing data corrupted by noise, it doesn't pose a challenge in an ideal environment [3];
- The value of the natural frequency, ω_n , proved to be just a scale factor in time, as will be demonstrated in Chapter 7 .

Change
chapter

The problem is then easily reduced to Equation (3.4):

$$G(s) = \frac{1}{1 + 2\zeta \left(\frac{s}{\omega_n}\right)^\nu + \left(\frac{s}{\omega_n}\right)^{\nu+1}} \quad (3.4)$$

Consequently, the first step will be to identify the fractional order, ν and the pseudo-damping coefficient, ζ , from the shape of the curve and then the natural frequency ω_n from the time values of the output.

This identification strategy follows the same rationale established by previous studies [3–5], which successfully demonstrated that the parameters of given fractional-order plants can be accurately estimated by mapping geometric regularities in their reaction curves. While those works focused on Fractional First-Order and Second-Species Commensurate Order systems, this thesis extends that methodology to the non-commensurate model described in eq. (3.2) and aims to validate whether the correlation between time domain features (such as $t_{0.5}$ and M_p) and system parameters remains robust enough for this more complex fractional-order structure.

4

This is the Fourth Chapter

5

This is the Fifth Chapter

6

Conclusion

7

Testing stuff

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do this and
that

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In this other case, another co-author is making a note about the citation for missing some bibliographic

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record [?, ?, ?].

Please note that the **FIRST LINE** of a paragraph after a **HEADING** (of Chapter or Section) **SHOULD NOT be indented**, as this is a typographical rule for this type of document.

Another rule is that you **SHOULD NEVER force LINE BREAKS** between paragraphs. You just need to leave a 'blank' line to separate paragraphs so that the document processor "understands" that the previous paragraph (independently of the number of "sentences" in it) was ended, and a new paragraphs is to be started. Note also that each paragraph is identified by a single LINE NUMBER in the editor pane.

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3G/4G Wireless Wide Area Networks (WWANs).

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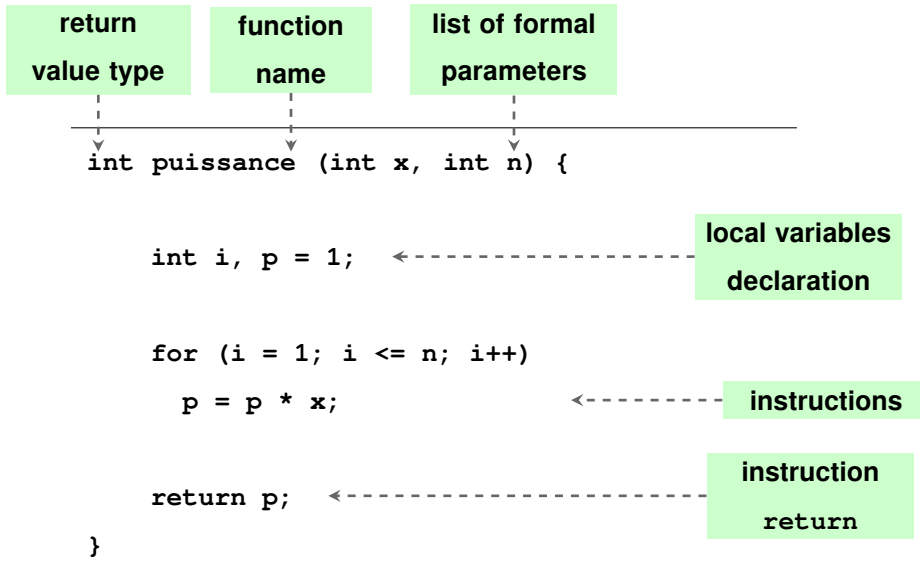
RC
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SYMBOL
defined in
"Glossary"

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Notice that mathematics makes extensive use of Formulas which are particularly well rendered in documents produced with LaTeX.

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ing of the
float object,
Figure

RC
example
of use of
Glossaries



Listing 7.1: A listing with a Tikz picture overlaid

Algorithm 7.1: Time Control Strategy

```

begin
    nextBitrate  $\leftarrow$  nextDownloadLevel
    nextBitrate  $\leftarrow$  GetNextBitrate()
    cpuLoad  $\leftarrow$  GetCpuLoad()
    bitrateDelta  $\leftarrow$  getBitrateDelta(currentBitrate, nextBitrate)
    if bitrateDelta > maxThreshold then
        SetBitrate(nextBitrate)
    if minThreshold < bitrateDelta < maxThreshold and numAttempts < 2 then
        numAttempts  $\leftarrow$  numAttempts + 1
    else if minThreshold < bitrateDelta < maxThreshold and numAttempts = 2 then
        numAttempts  $\leftarrow$  0
    else
        SetBitrate(nextBitrate)
    if 0 < bitrateDelta < minThreshold and numAttempts < 3 then
        numAttempts  $\leftarrow$  numAttempts + 1
    else if 0 < bitrateDelta < minThreshold and numAttempts = 3 then
        SetBitrate(nextBitrate)

```

Table 7.1 is automatically compressed to fit text width. You can use <https://www.tablesgenerator.com> to produce these tables, and then copy the $\text{\LaTeX}$ code generated to paste in the document.

Table 7.1: Comparison between today's and target Architectures of Telcos

Today		Target	
Rigid	Each evolutionary requirement involves development of multiple components, interfaces, platforms, etc.	Flexible	It is possible to modify or add new functionalities rapidly.
Slow	Development of a new application takes months or years.	Fast	Development of a new application takes weeks instead of months or years.
Closed	Limited integration with external environments.	Open	It is simple to integrate internal, applications with external entities.
Complex	Heterogeneous technologies, obsolescence, lack, of standards, high redundancy.	Standardised	Use of homogeneous architectural models.
Expensive	High Capex (for new service development) and, high, Opex (to ensure running of IT).	Cost-Effective	Capex and Opex are optimised.

Bibliography

- [1] A. Ben Hmed, M. Amairi, and M. Aoun, “Stability and resonance conditions of the non-commensurate elementary fractional transfer functions of the second kind,” *Communications in Nonlinear Science and Numerical Simulation*, vol. 22, no. 1-3, pp. 842–865, May 2015.
- [2] D. Valerio, S. Victor, and R. Malti, “Stability, resonance, and frequency response identification of second species fractional transfer functions.”
- [3] —, “Identification of a second-species commensurate fractional order transfer function from a step response with overshoot or oscillations.”
- [4] J. J. Gude, P. García Bringas, M. Herrera, L. Rincón, A. Di Teodoro, and O. Camacho, “Fractional-order model identification based on the process reaction curve: A unified framework for chemical processes,” *Results in Engineering*, vol. 21, p. 101757, Mar. 2024.
- [5] J. J. Gude and P. García Bringas, “Proposal of a General Identification Method for Fractional-Order Processes Based on the Process Reaction Curve,” *Fractal and Fractional*, vol. 6, no. 9, p. 526, Sep. 2022.

List of Symbols

A_s	Surface Area	μm^2
D_0	Initial Diameter	μm
ν	Real-valued Laplace variable exponent	
ω_n	Natural Frequency	rad/s
ζ	Pseudo-damping Coefficient	

Glossary

formula

A mathematical expression 12

LaTeX

LaTeX It is a mark up language specially suited for scientific documents as it can correctly format documents with all the typographical rules 12

mathematics

Mathematics is what mathematicians do 12



Code of Project

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

Listing A.1: Example of a XML file.

```
<?xml version="1.0" encoding="UTF-8"?>
<StreamInfo version="2.0">
  <Clip duration="PT01M0.00S">
    <BaseURL>videos/</BaseURL>
    <Description>svc_1</Description>
    <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="401.90"
      width="176" height="144" id="L0">
      <BaseURL>svc_1/</BaseURL>
      <SegmentInfo from="0" to="11" duration="PT5.00S">
        <BaseURL>svc_1-L0-</BaseURL>
      </SegmentInfo>
    </Representation>
    <Representation mimeType="video/SVC" codecs="svc" frameRate="30.00" bandwidth="1322.60"
      width="352" height="288" id="L1">
      <BaseURL>svc_1/</BaseURL>
      <SegmentInfo from="0" to="11" duration="PT5.00S">
```

```

        <BaseURL>svc_1-L1-</BaseURL>

    </SegmentInfo>

</Representation>

</Clip>

</StreamInfo>

```

Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Maecenas tincidunt velit quis orci. Sed in dui. Nullam ut mauris eu mi mollis luctus. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed cursus cursus velit. Sed a massa. Duis dignissim euismod quam.

Listing A.2: Assembler Main Code.

```

1  ; *****
2  ; * Constantes
3  ; *****
4
5  ON      EQU 1 ; contagem ligada
6  OFF     EQU 0 ; contagem desligada
7  INPUT   EQU 8000H ; endereco do porto de entrada
8           ; (bit 0 = RTC; bit 1 = botao)
9  OUTPUT  EQU 8000H ; endereco do porto de saida.
10
11
12 ; *****
13 ; * Stack
14 ; *****
15
16 PLACE   1000H
17 pilha:  TABLE 100H ; espaco reservado para a pilha
18 fim_pilha:
19
20 ; *****
21
22 PLACE   2000H
23
24 ; Tabela de vectores de interrupcao
25
26 tab:    WORD  rot0
27
28 ; *****
29 ; * Programa Principal
30 ; *****
31
32 PLACE   0
33
34 inicio:
35     MOV BTE, tab ; incializa BTE
36     MOV R9, INPUT ; endereco do porto de entrada
37     MOV R10, OUTPUT ; endereco do porto de saida
38     MOV SP, fim_pilha
39     MOV R5, 1 ; inicializa estado do processo P1
40     MOV R6, 1 ; inicializa estado do processo P2
41     MOV R4, OFF ; inicializa controle de RTC
42     MOV R8, 0 ; inicializa contador
43     MOV R7, OFF ; inicialmente nao permite contagem
44     EIO ; permite interrupcoes tipo 0
45     EI ; activa interrupcoes
46
47 ciclo:
48     CALL P1 ; invoca processo P1
49     CALL P2 ; invoca processo P2
50     JMP ciclo ; repete ciclo
51
52 ; *****
53 ; * ROTINAS
54 ; *****
55
56 P1:
57     CMP R5, 1 ; se estado = 1
58     JZ P1_1
59     CMP R5, 2 ; se estado = 2

```

```

60     JZ  P1_2
61 sai_P1:
62     RET          ; sai do processo.
63
64
65 P1_1:
66     MOVB R0, [R9] ; ler porto de entrada
67     BIT  R0, 1
68     JZ  sai_P1     ; se botao nao carregado, sai do processo
69     MOV R7, ON     ; permite contagem do display
70     MOV R5, 2      ; passa ao estado 2 do P1
71     JMP sai_P1
72
73 P1_2:
74     MOVB R0, [R9] ; ler porto de entrada
75     BIT  R0, 1
76     JNZ sai_P1     ; se botao continua carregado, sai do processo
77     MOV R7, OFF    ; caso contrario, desliga contagem do display
78     MOV R5, 1      ; passa ao estado 1 do P1
79     JMP sai_P1

```

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Listing A.3: HTML with CSS Code

```

1  <!DOCTYPE html>
2  <html>
3      <head>
4          <title>Listings Style Test</title>
5          <meta charset="UTF-8">
6          <style>
7              /* CSS Test */
8              * {
9                  padding: 0;
10                 border: 0;
11                 margin: 0;
12             }
13         </style>
14         <link rel="stylesheet" href="css/style.css" />
15     </head>
16     <header> hey </header>
17     <article> this is a article </article>
18     <body>
19         <!-- Paragraphs are fine -->

```

```

20 <div id="box">
21   <p>
22     Hello World
23   </p>
24   <p>Hello World</p>
25   <p id="test">Hello World</p>
26   <p></p>
27 </div>
28 <div>Test</div>
29 <!-- HTML script is not consistent -->
30 <script src="js/benchmark.js"></script>
31 <script>
32   function createSquare(x, y) {
33     // This is a comment.
34     var square = document.createElement('div');
35     square.style.width = square.style.height = '50px';
36     square.style.backgroundColor = 'blue';
37
38     /*
39      * This is another comment.
40      */
41     square.style.position = 'absolute';
42     square.style.left = x + 'px';
43     square.style.top = y + 'px';
44
45     var body = document.getElementsByTagName('body')[0];
46     body.appendChild(square);
47   };
48
49   // Please take a look at +=
50   window.addEventListener('mousedown', function(event) {
51     // German umlaut test: Berührungspunkt ermitteln
52     var x = event.touches[0].pageX;
53     var y = event.touches[0].pageY;
54     var lookAtThis += 1;
55   });
56 </script>
57 </body>

```

58 `</html>`

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

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Listing A.4: HTML CSS Javascript Code

```
1
2 @media only screen and (min-width: 768px) and (max-width: 991px) {
3
4     #main {
5         width: 712px;
6         padding: 100px 28px 120px;
7     }
8
9     /* .mono {
10         font-size: 90%;
11     } */
12
13     .cssbtn a {
14         margin-top: 10px;
15         margin-bottom: 10px;
16         width: 60px;
17         height: 60px;
18         font-size: 28px;
19         line-height: 62px;
20     }
```

Nulla dui purus, eleifend vel, consequat non, dictum porta, nulla. Duis ante mi, laoreet ut, commodo eleifend, cursus nec, lorem. Aenean eu est. Etiam imperdiet turpis. Praesent nec augue. Curabitur ligula quam, rutrum id, tempor sed, consequat ac, dui. Vestibulum accumsan eros nec magna. Vestibulum vitae dui. Vestibulum nec ligula et lorem consequat ullamcorper.

Listing A.5: PYTHON Code

```
1 class TelegramRequestHandler(object):
2     def handle(self):
3         addr = self.client_address[0]          # Client IP-adress
4         telegram = self.request.recv(1024)     # Recieve telegram
5         print "From: %s, Received: %s" % (addr, telegram)
6         return
```



A Large Table

Aliquam et nisl vel ligula consectetur suscipit. Morbi euismod enim eget neque. Donec sagittis massa. Vestibulum quis augue sit amet ipsum laoreet pretium. Nulla facilisi. Duis tincidunt, felis et luctus placerat, ipsum libero vestibulum sem, vitae elementum wisi ipsum a metus. Nulla a enim sed dui hendrerit lobortis. Donec lacinia vulputate magna. Vivamus suscipit lectus at quam. In lectus est, viverra a, ultricies ut, pulvinar vitae, tellus. Donec et lectus et sem rutrum sodales. Morbi cursus. Aliquam a odio. Sed tortor velit, convallis eget, porta interdum, convallis sed, tortor. Phasellus ac libero a lorem auctor mattis. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Nunc auctor bibendum eros. Maecenas porta accumsan mauris. Etiam enim enim, elementum sed, bibendum quis, rhoncus non, metus. Fusce neque dolor, adipiscing sed, consectetur et, lacinia sit amet, quam. Suspendisse wisi quam, consectetur in, blandit sed, suscipit eu, eros. Etiam ligula enim, tempor ut, blandit nec, mollis eu, lectus. Nam cursus. Vivamus iaculis. Aenean risus purus, pharetra in, blandit quis, gravida a, turpis. Donec nisl. Aenean eget mi. Fusce mattis est id diam. Phasellus faucibus interdum sapien. Duis quis nunc. Sed enim. Nunc auctor bibendum eros. Maecenas porta accumsan mauris. Etiam enim enim, elementum sed, bibendum quis, rhoncus non, metus. Fusce neque dolor, adipiscing sed, consectetur et, lacinia sit amet, quam.

As Table B.1 shows, the data can be inserted from a file, in the case of a somehow complex structure. Notice the Table footnotes.

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Table B.1: Example table

Benchmark: ANN	#Layers (1)	#Nets (2)	#Nodes* (3) = 8 · (1) · (2)	Critical path (4) = 4 · (1)	Latency (T_{iter}) (5)
A1	3–1501	1	24–12008	12–6004	4
A2	501	1	4008	2004	2–2000
A3	10	2–1024	160–81920	40	60 [†]
A4	10	50	4000	40	80–1200
Benchmark: FFT	FFT size [‡] (1)	#Inputs (2) = 2 ⁽¹⁾	#Nodes* (3) = 10 · (1) · (2)	Critical path (4) = 4 · (1)	Latency (T_{iter}) (5)
F1	1–10	2–1024	20–102400	4–40	6–60 [†]
F2	5	32	1600	20	40 – 1500
Benchmark: Random networks	#Types (1)	#Nodes (2)	#Networks (3)	Critical path (4)	Latency (T_{iter}) (5)
R1	3	10–2000	500	<i>variable</i>	(4)
R2	3	50	500	<i>variable</i>	(4) × [1; ⋯ ; 20]

* Excluding constant nodes.

[†] Value kept proportional to the critical path: (5) = (4) * 1.5.

[‡] A size of x corresponds to a 2^x point FFT.

Values in bold indicate the parameter being varied.

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And now an example (Table B.2) of a table that extends to more than one page. Notice the repetition of the Caption (with indication that is continued) and of the Header, as well as the continuation text at the bottom.

Table B.2: Example of a very long table spreading in several pages

Time (s)	Triple chosen	Other feasible triples
0	(1, 11, 13725)	(1, 12, 10980), (1, 13, 8235), (2, 2, 0), (3, 1, 0)
2745	(1, 12, 10980)	(1, 13, 8235), (2, 2, 0), (2, 3, 0), (3, 1, 0)
5490	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
8235	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
10980	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
13725	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
16470	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
19215	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
21960	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
24705	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
27450	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
30195	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
32940	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
Continued on next page...		

Table B.2: (...continued from previous page)

Time (s)	Triple chosen	Other feasible triples
35685	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
38430	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
41175	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
43920	(1, 13, 10980)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
46665	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
49410	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
52155	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
54900	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
57645	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
60390	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
63135	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
65880	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
68625	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
71370	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
74115	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
76860	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
79605	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
82350	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
85095	(1, 12, 13725)	(1, 13, 10980), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
87840	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
90585	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
93330	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
96075	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
98820	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
101565	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
104310	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
107055	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
109800	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
112545	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
115290	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
118035	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
120780	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
123525	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
126270	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
129015	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
131760	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
134505	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
137250	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
139995	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
142740	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
145485	(1, 12, 16470)	(1, 13, 13725), (2, 2, 2745), (2, 3, 0), (3, 1, 0)
148230	(2, 2, 2745)	(2, 3, 0), (3, 1, 0)
150975	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
153720	(1, 12, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
156465	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
159210	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
161955	(1, 13, 16470)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)
164700	(1, 13, 13725)	(2, 2, 2745), (2, 3, 0), (3, 1, 0)

An example of a large Table that autofits the size to the page margins is illustrated in Table B.3. Please notice the text size that is shrunken in order for the table to adjust to the page:

Table B.3: Sample Table.

URL	First Time Visit	Last Time Visit	URL Counts	Value	Reference
https://web.facebook.com/	1521241972	1522351859	177	56640	[facebook-2021]
http://localhost/phpmyadmin/	1518413861	1522075694	24	39312	database-management
https://mail.google.com/mail/u/	1516596003	1522352010	36	33264	Google-Gmail-2021
https://github.com/shawon100	1517215489	1522352266	37	27528	Code-Repository
https://www.youtube.com/	1517229227	1521978502	24	14792	Youtube-video-2021